



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	WEB TECHNOLOGY AND APPLICATION
COURSE CODE	:	ICT600
EXAMINATION	:	JULY 2024
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts :
PART A (5 Questions)
PART B (2 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages

PART A**QUESTION 1**

- a) Identify **FOUR (4)** applications of HTML language.

(8 marks)

- b) Identify **FIVE (5)** commonly used web-based Content Management Systems tools that are available.

(10 marks)

QUESTION 2

- a) Describe **SIX (6)** components of website anatomy.

(6 marks)

- b) Name **SIX (6)** development tools that can be used in client-side web app development.

(6 marks)

QUESTION 3

MySQL is one of the popular and widely used Relational Database Management System among most web developers which uses SQL as language for communicate with the system.

- a) Classify **FOUR (4)** of the mechanisms to enforce data restriction in database.

(8 marks)

- b) Produce an SQL statement to insert the following records into Table 1 named ITEM:

Table 1

itemID	itemBrand	itemType	itemPrice	quantity
sh234	Adidas	Shoe	345.00	2
ts124	Nike	T-Shirt	99.99	6
cp456	Puma	Cap	35.75	10

(10 marks)

QUESTION 4

- a) Produce Table 2 using PHP multi-dimensional array.

Table 2

Name of Car	Stock	Sold
Proton	15	8
Perodua	20	10
Toyota	10	5

(4 marks)

- b) Prepare the output of the following Php script.

```
<?php
$text = "Covid19 is a Disease Outbreak News";
echo strlen($text). "<br>";
$text = trim($text);
echo $text. "<br>";
echo strtoupper($text). "<br>";
echo strtolower($text). "<br>";
$text = str_replace("News", "Story", $text);
echo $text. "<br>";
echo substr($text, 2). "<br>";
var_dump(strpos($text, "Outbreak"));
echo "<br>";
var_dump(strpos($text, "News"));
?>
<?php
```

(6 marks)

QUESTION 5

The following is login script in PHP language.

```
<h1>Login</h1>

<form method="POST" action="login.php">
    <label for="username">User name:</label><br>
    <input id="username" name="username" type="text"> <br><br>

    <label for="password">Password: </label><br>
    <input id="password" name="password" type="password"> <br> <br>

    <input type="submit" value="Login" name="login">
</form>
<?php
if (isset ( $_POST ['login'] )) {
    $username = htmlentities($_POST ['username']);
    $password = htmlentities($_POST ['password']);

    $conn = mysqli_connect ("127.0.0.1", "root", "abcde", "ict600_db" ) or die ( mysqli_error () );

    // SQL query
    $strSQL = "SELECT `role`, `status`
                FROM `app user`
                WHERE username = '$username' AND password = PASSWORD('$password')";

    // Execute the query (the recordset $rs contains the result)
    $rs = mysqli_query ( $conn, $strSQL );
    $row = mysqli_fetch_array ( $rs, MYSQLI_ASSOC );
    if (mysqli_num_rows ( $rs ) == 1) {
        $_SESSION ['username'] = $username;
        $_SESSION ['role'] = $row ['role'];
        echo '<p>Hello ' . $username . '</p>';
        echo '<p>You are ' . $_SESSION ['role'] . ' user</p>';
        echo '<form method="get" action="index.php?page=home">';
        echo '<input type="submit" value="OK">';
        echo '</form>';
    } else {
        echo 'incorrect username and/or password. Please login again.';
    }
} else {
    echo '<p>Hello please login</p>';
    $username = '';
    $_SESSION ['username'] = '';
}
}
```

- a) Produce a partial PHP script to check that only the administrator role is allowed to open the user_view.php page otherwise, the user will be directed to an error page (6 marks)
- b) Produce a PHP script to log out when a user clicks the logout hyperlink. Assume that each has a logout hyperlink as the following.

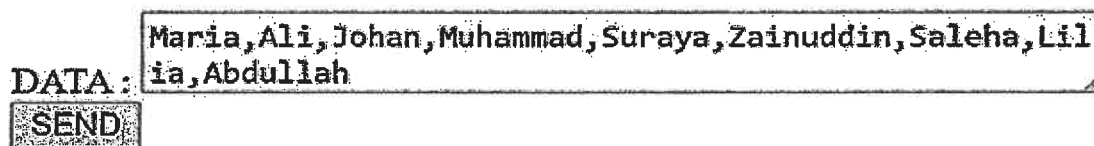
```
<a href="index.php?page=home&logout"> Logout</a>
```

(6 marks)

PART B**QUESTION 1**

Department of Analytics wants you to develop a simple web application that can be used to analyze the length of patient's name. You are required to determine on how to develop **TWO (2)** pages based on description below:

1. First Page (inputData.html) that is used to input the data which is the list of patient's name that is separated by commas (see the Figure 1). Then, this page will be submitted to receiveData.php



DATA : Maria, Ali, Johan, Muhammad, Suraya, Zainuddin, Saleha, Lilia, Abdullah

SEND

Figure 1: inputData.html

2. The second page (receiveData.php) will receive the data (list of patient's name) from the previous page (inputData.html). By using the received data, this page will separate the name into an index array. This page will create the associative array named \$dataArray where the key is patient's name and the value is the length of patient name. From the data of \$dataArray, this page will find and display the key and the value of patient with the longest name.

(15 marks)

QUESTION 2

You are assigned to recommend the construction of a user registration page for Kelab Usahawan KPPIM website using PHP as shown in Figure 2. User will need to enter user profile information and password. The user must enter their full name, email address, and password. Other optional information is birthdate and gender. When the user enters all required information, a successful message will appear on the page. Otherwise, an unsuccessful error message will appear.

Note: The information does not need to be stored in the database.

User Signup

*** required informed Fullname***

email*

Password*

Birthdate

Gender:

☒ Male ☐ Female

User signup Form

User Signup

Fullname : Muhammad Ali

Email : mali@gmail.com

Gender : M

Birthdate : 1989-03-23

Thank you. You are successfully signup.

Successful Signup Message

User Signup

One or more required information are not entered.

Please enter all the required information.

Unsuccessful Signup Message

Figure 2 : User registration page for Kelab Usahawan KPPIM

(15 marks)

END OF QUESTION PAPER